

JOUR 8500 RLA

Productivity, prominence, and the effects of academic environment

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Way, S. F., Morgan, A. C., Larremore, D. B., & Clauset, A. (2019). Productivity, prominence, and the effects of academic environment. *Proceedings of the National Academy of Sciences*, 116(22), 10729-10733.

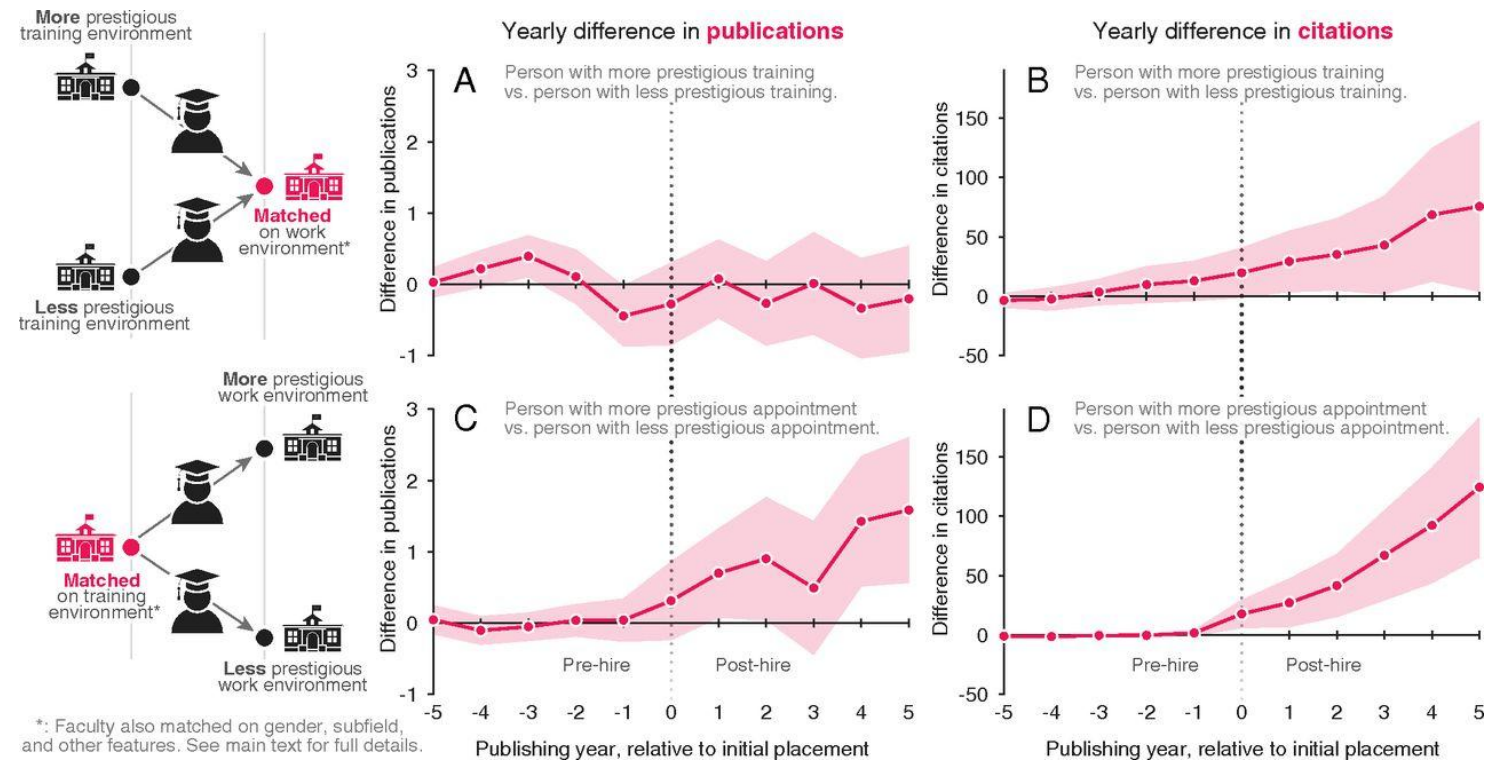
Background

- **Faculty at more-prestigious institutions produce:**
 - More scientific literature
 - Receive more citations and scientific awards
 - Train more faculty hired by other prestigious institutions
- For productive scholars, which of **where they trained** and **where they work** explains their academic performance (number of publications and citations)?

Methods

- **quasi-natural experiment**
(discontinuity: institution moving)
- **Dataset:**
 - 1970-2011
 - 2,453 tenure-track faculty
 - 205 CS departments
 - more than 200,000 publications and 7.4 million citations

FIG 2

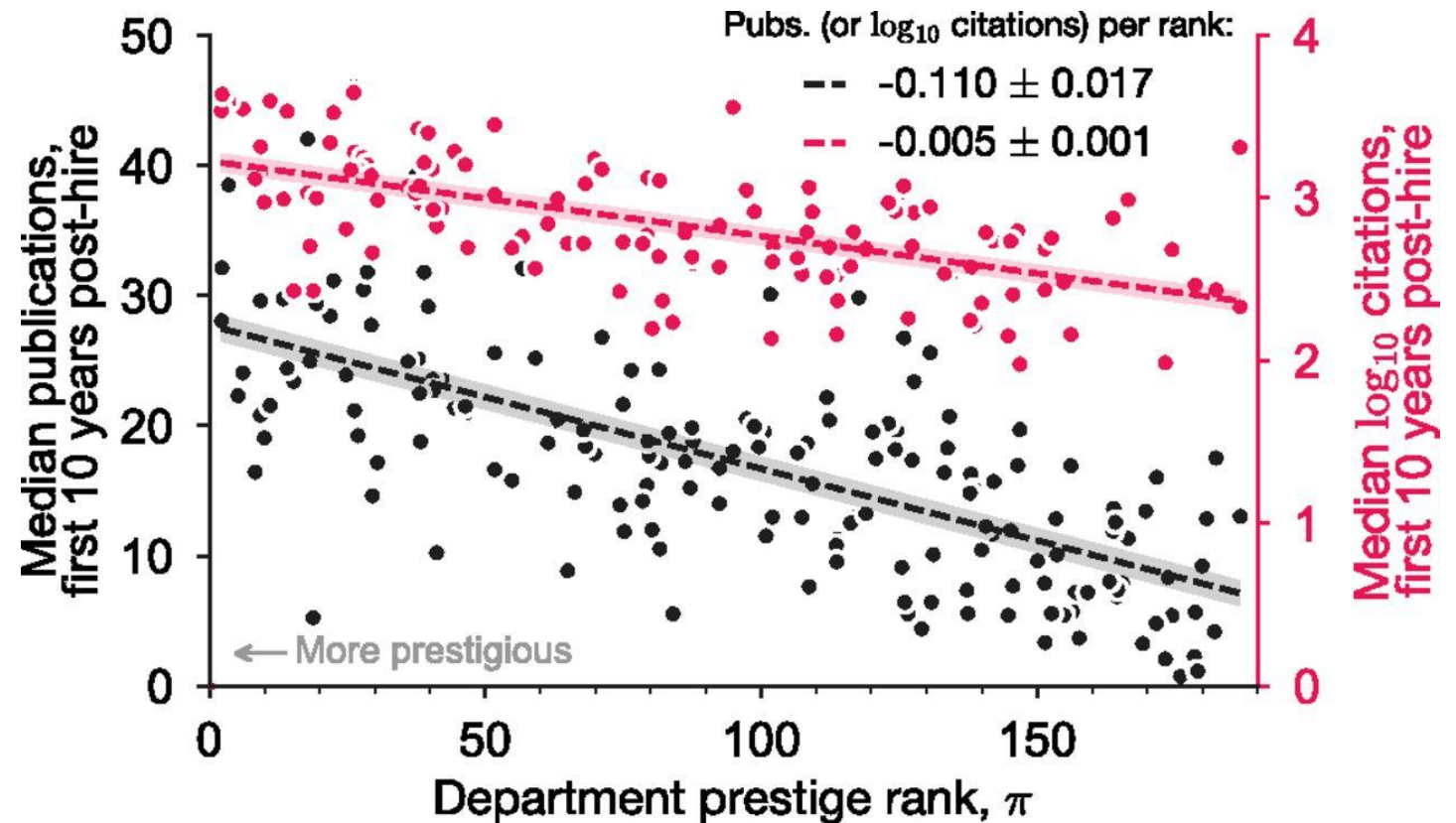


Early-career productivity is driven by work environment prestige. For pairs of computer science faculty matched by (A and B) work environment prestige or (C and D) training environment prestige, (A) publication and (B) citation counts are statistically independent of differences in doctoral prestige but are driven higher by (C and D) placing into a more prestigious work environment. Shaded regions denote 95% confidence intervals for the mean. Similar results are obtained using U.S. News & World Report department rankings in place of prestige (see [SI Appendix, Fig. S1](#)).

Results

- More prestigious training
 - not more productive
 - more citations
- More prestigious appointment
 - More papers (5.1 more papers in the first 5 years)
 - More citations (344 more)
- Conclusion: appointment instead of training explains academic output

FIG 1



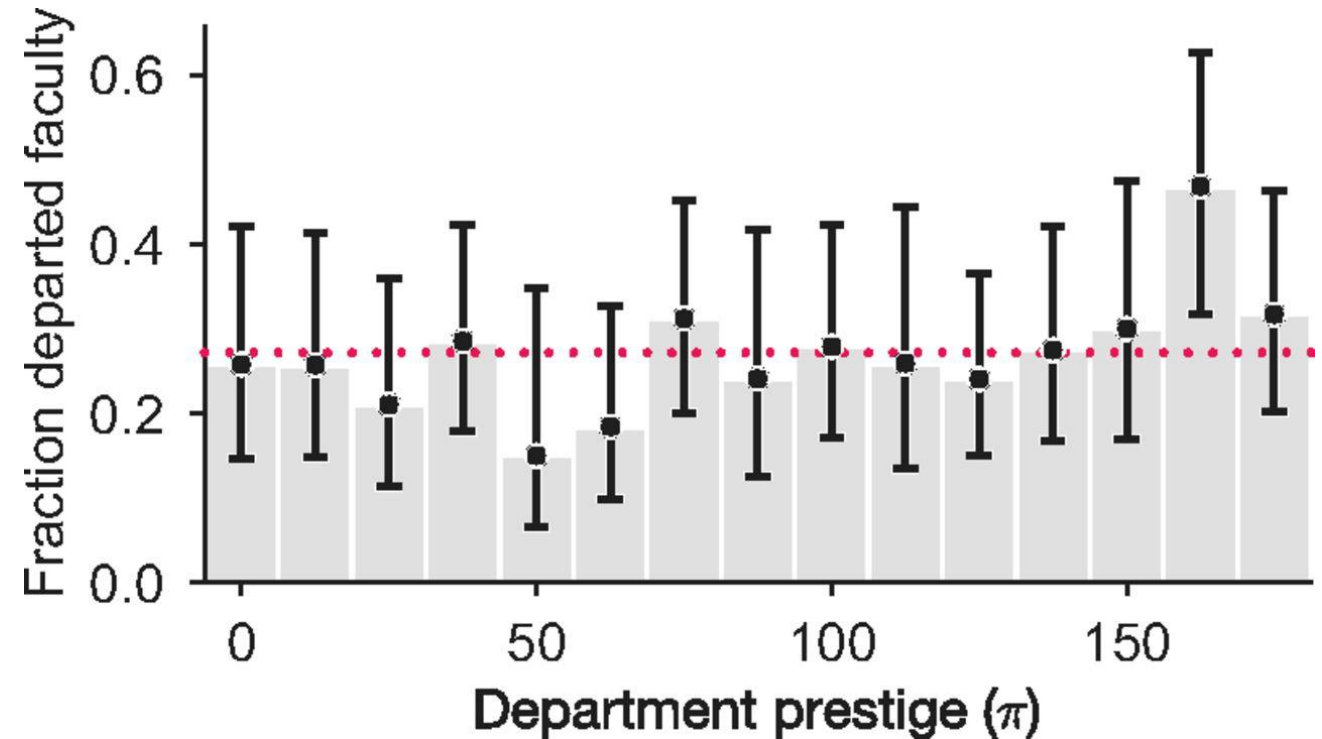
Institutional prestige predicts early-career productivity and prominence of computer science faculty. Shown are median publication (left axis) and $\log_{10}$ citation (right axis) counts per faculty per institution (minimum three faculty per institution), accumulated through their first 10 years posthire, adjusted for growth in publication rates over time ([SI Appendix, section A](#)). Shaded regions denote 95% confidence intervals for least squares regression.

Results

- **Retention**

- Retention rate (71.9%)
- productivity alone is a weak prediction of retention
- selective retention of more-productive individuals cannot explain the magnitude of the prestige–productivity effect

FIG 3

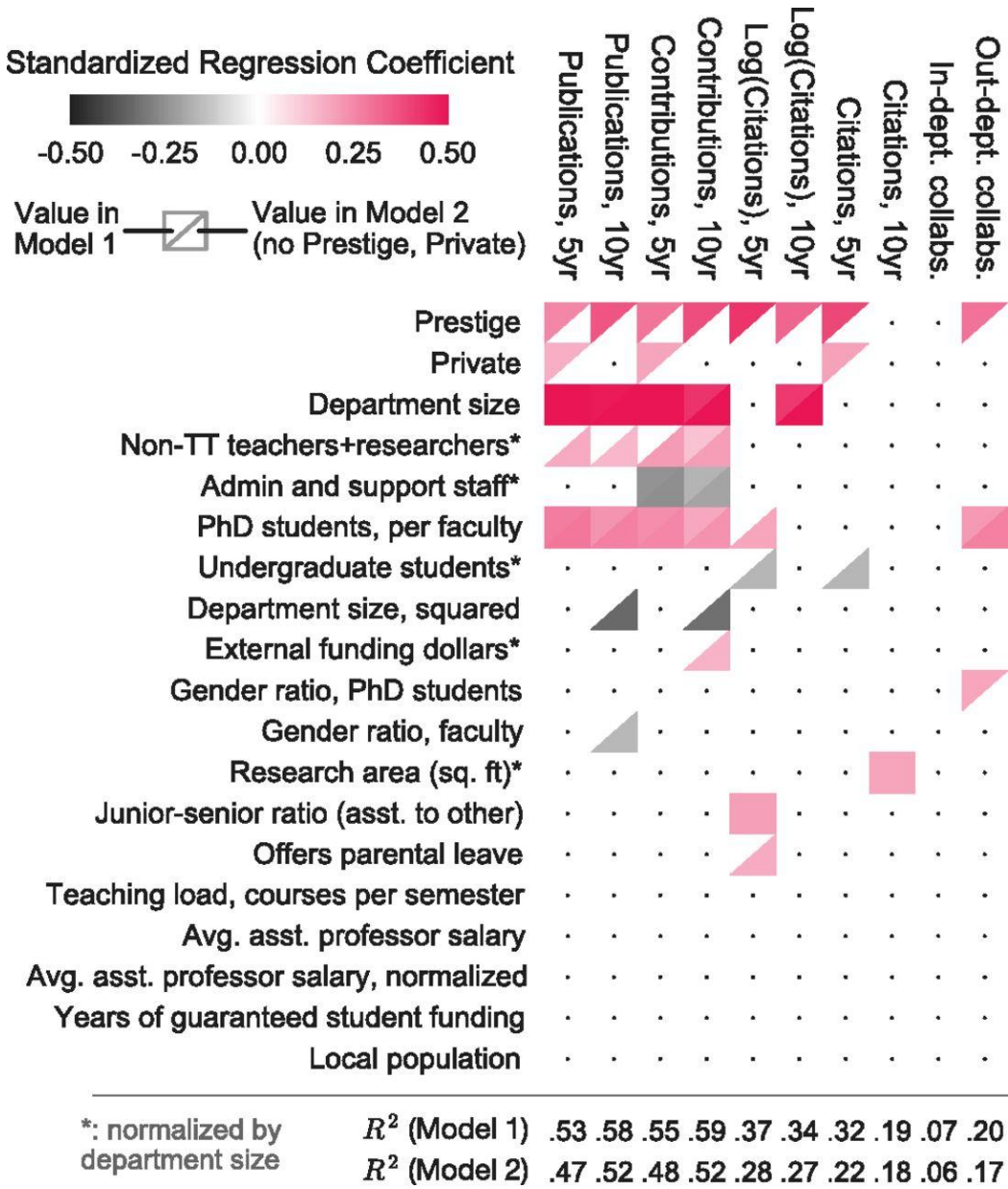


Binned by departmental prestige, early-career faculty leave their initial appointments, either for other institutions or for a nonacademic job, at similar rates. Error bars indicate 95% confidence intervals around the means. Dotted magenta line indicates the total fraction of departed faculty from all 205 institutions (0.281).

Results

- Departmental facilitation and its variation with prestige
 - Focus on early-career individuals, does not necessarily contradict past studies
 - Few correlation between most departmental characteristics and scholarly output

FIG 4



Discussion

- **For early-career faculty, the characteristics of their working environment rather than doctoral training, drive their productivity**
 - preferential selection or retention do not explain this
- **Cumulative advantage:**
 - Prestige in the past predict prestige after graduation
- **Contribute to the assumption that meritocratic principles or mechanisms govern the production of knowledge**

Questions

- **Figure 1.**

- Are there risks of reader misunderstanding or misinterpretation in a dual-axis chart such as this one? (Anastasia)
- Figure 1 only includes median publications. Whether there is a way to adjust this one by including variations? (Rachel)
- Is there a reason $\log_{10}$ is more commonly used in mass communication or computational methods papers? (Quin)
- Should Figure 1 add a legend about what different colors mean? The figure uses two y-axis, would it be possible to replace this figure with two sub-figures instead? (Jikai)

Questions

- **Figure 2.**
 - If using only difference scores might hide differences in absolute productivity levels. Also, since the analysis focuses on early-career years, could it be that the effect of doctoral training appears later, as scholars develop over time? (Carolina)
 - How the results in Figure 2 might differ if median differences were used instead of means? (Wenwen)
 - Would it improve readability if the visualization were designed so that this comparison is clearer even without reading text? (Eunsun)

Questions

- **Figure 3.**

- Figure 3 includes error bars that represent 95% confidence intervals. What does the Figure 3 communicate by showing error bars?(Namuun)

- **Figure 4.**

- In Figure 4, how does the heatmap's design allow for the simultaneous comparison of two different regression models within a single cell, and what visual cue is used to denote results that lack statistical significance? Would this have been better represented using tables? (Jessica)